Logarithmic and Exponential Functions

ANSWER KEY

Section 1: Fundamental Laws of Logarithms

1 Evaluate $\log_3(54) - \log_3(2)$ without a calculator.

a. 3 ✓

b. 2

c. 4

d. $\log_3(52)$

$\log_3(54/2) = \log_3(27) = 3$ since $3^3 = 27$.

2 Express $2\log_a(x) - 3\log_a(y) + \log_a(z)$ as a single logarithm.

a. $\log_a\left(\frac{x^2z}{y^3}\right)$ ✓

b. $\log_a\left(\frac{2xz}{3y}\right)$

c. $\log_a(x^2 - y^3 + z)$

d. $\log_a\left(\frac{x^2}{y^3z}\right)$

Power law: $\log_a(x^2) - \log_a(y^3) + \log_a(z) = \log_a\left(\frac{x^2z}{y^3}\right)$.

3 Given $\log_b(2) = p$ and $\log_b(3) = q$, express $\log_b(72)$ in terms of p and q.

a. $3p + 2q$ ✓

b. $2p + 3q$

c. $6pq$

d. $3p - 2q$

$72 = 2^3 \times 3^2 \Rightarrow \log_b(72) = 3\log_b(2) + 2\log_b(3) = 3p + 2q$.

Section 2: Solving Logarithmic Equations

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Solve the equation $\log_2(x) + \log_2(x-2) = 3$ for $x > 2$.

a. $x = 4$ ✓

b. $x = 4$ and $x = -2$

c. $x = 3$

d. $x = 8$

$\log_2(x(x-2)) = 3 \Rightarrow x^2 - 2x = 2^3 = 8 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x-4)(x+2) = 0$. Since $x > 2$, $x = 4$.

5 Solve $\log_3(2x+1) - \log_3(x-1) = 2$.

a. $x = \frac{10}{7}$ ✓

b. $x = 2$

c. $x = \frac{7}{10}$

d. No real solution

$\log_3\left(\frac{2x+1}{x-1}\right) = 2 \Rightarrow \frac{2x+1}{x-1} = 9 \Rightarrow 2x+1 = 9x-9 \Rightarrow 7x = 10 \Rightarrow x = 10/7$.

6 Solve the base-change logarithmic equation $\log_4(x) + \log_2(x) = 6$.

a. $x = 16$ ✓

b. $x = 8$

c. $x = 64$

d. $x = 4$

$\log_4(x) = \frac{\log_2(x)}{\log_2(4)} = \frac{1}{2}\log_2(x)$. Thus $\frac{3}{2}\log_2(x) = 6 \Rightarrow \log_2(x) = 4 \Rightarrow x = 2^4 = 16$.

Section 3: Exponential Equations and Growth Models

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 Solve $3^{2x-1} = 5^x$, giving x in terms of natural logarithms.

a. $x = \frac{\ln 3}{2\ln 3 - \ln 5}$ ✓

b. $x = \frac{\ln 5}{\ln 3}$

c. $x = \frac{\ln 3}{\ln 5 - 2\ln 3}$

d. $x = \frac{2\ln 3}{\ln 5}$

Take $\ln$: $(2x-1)\ln 3 = x\ln 5 \Rightarrow 2x\ln 3 - \ln 3 = x\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = \ln 3 \Rightarrow x = \frac{\ln 3}{2\ln 3 - \ln 5}$.

8 A bacterial culture grows according to $N(t) = 500e^{0.04t}$, where t is in hours. After how many hours will the population reach 2000?

a. $t = 25\ln 4 \approx 34.7$ hr ✓

b. $t = 50$ hr

c. $t = 20\ln 2 \approx 13.9$ hr

d. $t = 100$ hr

$500e^{0.04t} = 2000 \Rightarrow e^{0.04t} = 4 \Rightarrow 0.04t = \ln 4 \Rightarrow t = \frac{\ln 4}{0.04} = 25\ln 4$.

9 Solve the simultaneous exponential system $2^x 4^y = 32$ and $3^{x-y} = 27$.

a. $x = 4, y = \frac{1}{2}$ ✓

b. $x = 3, y = 1$

c. $x = 5, y = 2$

d. $x = 2, y = 1$

$2^{x+2y} = 2^5 \Rightarrow x + 2y = 5$. Also $3^{x-y} = 3^3 \Rightarrow x - y = 3$. Subtracting gives $3y = 2 \Rightarrow y = 1/2, x = 4$.

Section 4: Reduction to Linear Form ($Y = mX + c$)

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 The variables x and y are related by $y = ax^b$. When $\ln y$ is plotted against $\ln x$, a straight line passes through $(0, 2)$ with slope 3. Find a and b.

a. $a = e^2, \quad b = 3$ ✓

b. $a = 2, \quad b = 3$

c. $a = 3, \quad b = 2$

d. $a = e^3, \quad b = e^2$

$\ln y = \ln a + b\ln x$. Slope $m = b = 3$. Intercept $c = \ln a = 2 \Rightarrow a = e^2$.

11 The relation $y = Ab^x$ is linearized by plotting $\lg y$ against x. If the straight line has y-intercept 1 and gradient 0.3, find A and b.

a. $A = 10, \quad b = 10^{0.3}$ ✓

b. $A = 1, \quad b = 0.3$

c. $A = 10^{0.3}, \quad b = 10$

d. $A = 0.3, \quad b = 10$

$\lg y = \lg A + x\lg b$. Intercept $\lg A = 1 \Rightarrow A = 10$. Gradient $\lg b = 0.3 \Rightarrow b = 10^{0.3}$.

12 If a graph of xy against x^2 is a straight line with slope 4 and vertical intercept -5, express y in terms of x.

a. $y = 4x - \frac{5}{x}$ ✓

b. $y = 4x^2 - 5$

c. $y = 4 - \frac{5}{x^2}$

d. $y = \frac{4}{x} - 5$

$xy = 4(x^2) - 5 \Rightarrow y = 4x - \frac{5}{x}$.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647